$$\begin{pmatrix} e_{1t} \\ e_{2t} \end{pmatrix} = \begin{pmatrix} \varepsilon_{1t} - b_{12}\varepsilon_{2t} \\ \varepsilon_{2t} \end{pmatrix} \sim \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}; \begin{pmatrix} 1+4b_{12}^2 & -b_{12}\cdot 4 \\ -b_{12}\cdot 4 & 4 \end{pmatrix} \right)$$

$$D(\varepsilon_{1t} - b_{12}\varepsilon_{2t}) = D\varepsilon_{1t} + b_{12}^2 D\varepsilon_{2t} + 2(-b_{12})\,cov(\varepsilon_{1t}, \varepsilon_{2t})$$

$$cov(\varepsilon_{1t} - b_{12}\varepsilon_{2t}, \varepsilon_{2t}) = E(\varepsilon_{1t}\varepsilon_{2t} - b_{12}\varepsilon_{2t}^2) = -b_{12}\cdot D\varepsilon_{2t}$$

29.11 Семинар

$$\begin{pmatrix} 1 & b_{12} \\ b_{21} & 1 \end{pmatrix}\begin{pmatrix} y_t \\ z_t \end{pmatrix} = \begin{pmatrix} \gamma_{10} \\ \gamma_{20} \end{pmatrix} + \begin{pmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{21} & \gamma_{22} \end{pmatrix}\begin{pmatrix} y_{t-1} \\ z_{t-1} \end{pmatrix} + \begin{pmatrix} \varepsilon_y \\ \varepsilon_z \end{pmatrix}$$

$\downarrow \cdot B^{-1}$ $\uparrow \cdot B$

$$\begin{pmatrix} y_t \\ z_t \end{pmatrix} = \begin{pmatrix} a_{10} \\ a_{20} \end{pmatrix} + \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}\begin{pmatrix} y_{t-1} \\ z_{t-1} \end{pmatrix} + \begin{pmatrix} e_y \\ e_z \end{pmatrix}$$

$$\Sigma_e = \begin{pmatrix} 0.6 & 0.3 \\ 0.3 & 0.5 \end{pmatrix}$$

$$\begin{pmatrix} \varepsilon_{yt} \\ \varepsilon_{zt} \end{pmatrix} = B\begin{pmatrix} e_{yt} \\ e_{zt} \end{pmatrix}$$

$$\Sigma_\varepsilon = B\cdot\Sigma_e\cdot B^T \Rightarrow \Sigma_e = B^{-1}\Sigma_{\varepsilon_t}(B^{-1})^T$$

$$\begin{pmatrix} D\varepsilon_1 & 0 \\ 0 & D\varepsilon_2 \end{pmatrix} = \begin{pmatrix} 1 & b_{12} \\ b_{21} & 1 \end{pmatrix} \begin{pmatrix} 0.6 & 0.3 \\ 0.3 & 0.5 \end{pmatrix} \begin{pmatrix} 1 & b_{21} \\ b_{12} & 1 \end{pmatrix} =$$

$$\begin{pmatrix} 0.6 + 0.6\,b_{12} + 0.5\,b_{12}^2 & \cdots \\ 0.6\,b_{21} + 0.3\,b_{21}b_{12} + 0.5\,b_2 + 0.3 & 0.5 + 0.6\,b_{21} + 0.6\,b_{21}^2 \end{pmatrix} =$$

$\Rightarrow b_{12} = 0$ по Чолески $\Rightarrow b_{21} = -0.5$

$$= \begin{pmatrix} 0.6 & 0.6\,b_{21} + 0.3 \\ 0.6\,b_{21} + 0.3 & \underbrace{0.5 + 0.6\,b_{21} + 0.6\,b_{21}^2}_{0.35} \end{pmatrix}$$

№2

Теперь $\begin{pmatrix} 0.7 & 0.3 \\ 0.3 & 0.2 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$

$$(1 - a_{11}L)(1 - a_{22}L) - a_{12}a_{21}L^2 = 0$$

$$(1 - 0.7L)(1 - 0.2L) - 0.3^2L^2 = 0$$

$L_1 = 1.25 > 1$

$L_2 = 10 > 1$ $\Rightarrow$ stability $\Rightarrow$ stationar.

$\det(I - \Sigma_\varepsilon \cdot L) = 0$

$|L_1| > 1 \Rightarrow$ stab. $\Rightarrow$ stat.

$|L_2|$

№3. Прод. задачу

$$\begin{pmatrix} 1 & b_{12} \\ b_{21} & 1 \end{pmatrix}\begin{pmatrix} y_t \\ z_t \end{pmatrix} = \begin{pmatrix} \gamma_{10} \\ \gamma_{20} \end{pmatrix} + \begin{pmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{21} & \gamma_{22} \end{pmatrix}\begin{pmatrix} y_{t-1} \\ z_{t-1} \end{pmatrix} + \begin{pmatrix} \varepsilon_y \\ \varepsilon_z \end{pmatrix}$$

$\downarrow \cdot B^{-1}$ $\uparrow \cdot B$

$$\begin{pmatrix} y_t \\ z_t \end{pmatrix} = \begin{pmatrix} a_{10} \\ a_{20} \end{pmatrix} + \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}\begin{pmatrix} y_{t-1} \\ z_{t-1} \end{pmatrix} + \begin{pmatrix} e_y \\ e_z \end{pmatrix}$$

$$\Sigma e = \begin{pmatrix} 0.6 & 0.3 \\ 0.3 & 0.5 \end{pmatrix}$$

$$\det\left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 0.6 & 0.3 \\ 0.3 & 0.5 \end{pmatrix}L\right) = \det\begin{pmatrix} 1-0.6L & -0.3L \\ -0.3L & 1-0.5L \end{pmatrix} =$$

$$= (1-0.6L)(1-0.5L) - 0.09L^2 = 0$$

$$1 - 1.1L + 0.3L^2 - 0.09L^2 = 0$$

$$1 - 1.1L + 0.21L^2 = 0$$

$L = 1.17 > 1$

$L = 4.06 > 1$

$\Rightarrow$ stab. $\Rightarrow$ stat.